5a	$+\Delta U$ increase in internal energy $+q$ heat energy transferred to the system $+w$ work done on system (1 error -1 mark)	B2
bi	$W = p\Delta V$ $= 5.2 \times 10^5 \times (5.0 - 1.6) \times 10^{-4}$ $= 177\ J$ $\Delta U = q + w$ $= 442 - 177$ $\Delta U = 265\ J$	M1 M1
bii	No molecular/microscopic potential energy. Internal energy decreases, so total molecular/random/microscopic kinetic energy decreases. Mean molecular kinetic energy is proportional to thermodynamic temperature and decreases, so temperature decreases.	B1 B1 B1
biii	$\Delta U + 265 - 313 = 0$ $\Delta U = 48$ J	A1
biv	$pV = NkT$ $5.2 \times 10^5 \times 1.6 \times 10^{-4} = N \times 1.38 \times 10^{-23} \times (273 + 227)$ $N = 1.2 \times 10^{22}$	C1 A1